Assignment 2

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Final Report : Campus Placement Prediction Using Machine Learning

1. **Dataset Selection:**
   This dataset was sourced from Kaggle:
   🔗 ML with Python Course Project (https://www.kaggle.com/c/ml-with-python-course project/data?select=train.csv)
   It contains academic and employment-related records of 215 students, aimed at predicting their placement status — the categorical target variable, "status" (Placed / Not Placed).
   Dataset Composition:

- Total Records: 215
- Initial Columns: 15
- Dropped Columns:
  - sl_no: A simple index with no predictive power.
  - salary: Dropped due to excessive missing values and potential target leakage (only available for placed students).

Final Dataset:

- Columns: 13
- Data Types: Mixture of categorical and numerical
- Missing Values: None after preprocessing

2. **Data Preprocessing:**

The preprocessing steps ensured the dataset was clean, standardized, and suitable for machine learning:

- **Dropped Irrelevant Columns:** sl_no and salary
- **Confirmed No Missing Values** in the remaining columns
- **Encoded Categorical Variables** using Label Encoding
- **Standardized Numerical Features** using StandardScaler
- **Split Dataset:** 70% training and 30% testing

3. Exploratory Data Analysis (EDA)

Key findings and visualizations from EDA:

- **Class Imbalance:** 148 students placed, 67 unplaced.
- **Distribution Plots:** Academic scores (e.g., `ssc_p`, `hsc_p`, `degree_p`, `etest_p`) were visualized using histograms and KDE plots.
- **Categorical Analysis:** Count plots and bar charts for features like `gender`, `workex`, `specialisation`, and `degree_t` showed their relationship with placement status.

- **Heatmap of Correlation:** Revealed weak-to-moderate relationships among numerical features and the target

4. Model Selection and Tuning

To ensure robust prediction, four models were selected for complementary strengths:

1. Logistic Regression

- Chosen as the baseline for its simplicity and interpretability
- Assumes a linear relationship between features and log-odds of placement

2. Random Forest Classifier

- Captures non-linear relationships and handles both types of features well
- Robust to outliers and noise, and useful for feature importance

3. Support Vector Machine (SVM)

- Effective in high-dimensional space
- Useful for complex boundaries using kernel tricks

4. Voting Classifier

- Aggregates predictions of Logistic Regression, Random Forest, and SVM
- Enhances generalization by combining diverse model behaviors using **hard voting**

✅ **Hyperparameter tuning** was conducted using **GridSearchCV with cross-validation**
✅ Evaluation metrics included: Accuracy, Precision, Recall, and F1-Score

5. Model Evaluation

Each model was evaluated on the test dataset. Below is the comparison of their tuned performance:

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**
Logistic Regression	83.08%	85.11%	90.91%	87.91%
Decision Tree	84.62%	86.96%	90.91%	88.89%
Random Forest	80.00%	79.25%	95.45%	86.60%
SVM	78.46%	77.78%	95.45%	85.71%
k-NN	73.85%	74.55%	93.18%	82.83%

- **Decision Tree** achieved the highest accuracy (84.62%) and F1 score (88.89%).
- **Logistic Regression** also showed strong balanced performance.
- **Random Forest** and **SVM** had excellent recall, useful when minimizing false negatives is important.
- **k-NN** performed moderately but had the lowest accuracy among the five.

6. Visualizations & Interpretations

- **Confusion Matrices** were plotted for each model to highlight TP, FP, TN, FN
- **Bar Plots & Heatmaps** were used to visualize performance metrics
- **Feature Importance Plot** was generated using the Random Forest model

- ## 7. Conclusion & Observations

  - ✓ Decision Tree emerged as the top performer based on accuracy and F1 score.
  - ✓ Logistic Regression provided a well-balanced performance across all metrics.
  - ✓ Random Forest and SVM are valuable for scenarios prioritizing recall, reducing false negatives.
  - ✓ k-NN is a decent baseline but may require further tuning or feature engineering.
  - ✓ Hyperparameter tuning significantly improved model performances.
  - ✓ Preprocessing steps, including encoding and scaling, played a critical role in achieving reliable results.
  - ✓ EDA insights on academic performance and categorical features helped understand placement factors.